OntarioTech 5785G Assignment: Sensitivity Analysis of a V2X Offloading Environment

Objective

The objective of this assignment is to investigate how different networking and computing parameters influence the response time experienced by users in a Vehicle-to-Everything (V2X) offloading system. Moreover, you will explore how changes in the multi-agent backbone architecture affect the learning performance of the agents in the system.

1 Part A: Resource Profile Analysis [20 Marks]

Consider the environment parameters, as found in the code, where task complexity is defined as 8e8 CPU cycles per MB, the Fog server capacity is 16 GHz, the Cloud capacity is 80 GHz, and the backhaul delay is 35ms: Modify the environment to create and evaluate the following deployment profiles. For each profile, write the four lines of codes that need to change, mention which function and in which python class these lines are located.

1. **Edge-Only Profile**

   Configure the environment such that all tasks are executed locally on the vehicle without using the RSU/Fog or Cloud layers.

2. **RSU/Fog-Only Profile**

   Configure the environment such that all tasks are offloaded to the RSU/Fog layer and neither local execution nor Cloud execution is allowed.

3. **Cloud-Only Profile**

   Configure the environment such that all tasks are offloaded to the Cloud and neither local execution nor Fog execution is allowed.

Question A1

Compare the three profiles from the user's perspective.

1. Which profile achieved the lowest average response time?
2. How do communication delays and computational resources contribute to the observed behavior?

2 Part B: Number of RSUs [10 Marks]

The current environment contains 2 RSUs: Modify the number of RSUs (write the line of code that is needed to change, in which function and in which python class)

Question B1

Does increasing the number of RSUs reduce overall latency?
Explain the relationship between communication distance and transmission rate.

3 Part C: Data Size Analysis [10 Marks]

The current task size range is 0.5 to 4.0 Change the code needed to increase the task size to a range between 1.0 and 8.0 MB (mention the line of code needed to change, in which function and in which python class)

Question C1

How does increasing task size affect latency?

4 Part D: Edge CPU Frequency Analysis [10 Marks]

The current edge CPU frequency is generated using a random uniform distribution between 1.0 and 2.5 GHz: Modify the code to increase the edge CPU frequency to a range between 2.0 and 4.0 GHz (mention the line of code needed to change, in which function and in which python class)

Question D1

How does increasing edge computational power affect response time?

5 Part E: Latency Component Analysis [15 Marks]

The current latency equations for the cloud is as follows:

$$t_{exec} = \frac{b}{r_{v2i}} + d_{backhaul} + \frac{b \cdot c_i}{f_{cloud}} \tag{1}$$

Where b is the task size in MB, r_{v2i} is the vehicle-to-infrastructure transmission rate in MB/s, $d_{backhaul}$ is the backhaul delay in seconds, c_i is the computational complexity in CPU cycles per MB, and f_{cloud} is the cloud CPU frequency in GHz.

Change the code to only include the communication delay component (i.e., remove the backhaul and cloud execution components), mention the line of code needed to change, in which function and in which python class.

6 Part F: Multi-Agent Backbone Change [35 Marks]

The current multi-agent backbone is implemented using

1. MLP
2. LSTM
3. Transformer

Add another backbone architecture using "FITS: Modeling Time Series with 10k Parameters" architecture. You first need to check the code from the paper, and use the same architecture to implement the new backbone.

Code to be added in __init__ function of the DecentralizedAgent class in the file "agents/madrl_agent.py":

Code to be added in the forward function of the DecentralizedAgent class in the file "agents/madrl_agent.py":

Submission

Submit only the completed PDF file with all answer boxes filled in.

- Do **not** submit any separate Python files, source code files, ZIP archives, screenshots, or additional documents.

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- For each modification requested in this assignment, include the relevant code changes directly within the corresponding answer box.
- When describing a code modification, clearly specify:
  1. The Python class in which the change was made.
  2. The function or method that was modified.
  3. The updated line(s) of code.

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